
pool all subjects & slides
cluster labels
stack S=2 slides
group lasso, λ by CV
predict
g = 1
g = 2
slide s=1
slide s=2
q
n subjects
S
eGFR
subjects
λ by 5-fold CV
Biopsy Slide
K tubule objects
GMM Clustering
G=2, pooled across subjects & slides
Feature Extraction
(G×q) per slide, per subject
Tensor Assembly
$\mathcal{X} \in \mathbb{R}^{G \times q \times S \times n}$
Group Lasso Fit
$\hat{\mathbf{B}} \in \mathbb{R}^{G \times q \times S}$, sparse
Predicted eGFR
$\hat{y}_i = \hat{\mu} + \langle \mathbf{B}, \mathbf{X}_i \rangle_F$
Cluster 1 (non-diseased)
Cluster 2 (diseased)
Selected block ($\hat{\mathbf{B}}_{\cdot,j_\ell,\cdot} \neq \mathbf{0}$)
Zeroed block ($\hat{\mathbf{B}}_{\cdot,j_\ell,\cdot} = \mathbf{0}$)